

CliniCalm: An Intake Quality Gate for Clinical Trial Data

Product Requirements Document | September 18, 2026 | POC: Anvitha Abbagani

B2B SaaS concept, developed as an independent case study against Flatiron Health's market — not an official Flatiron product. [Try the live prototype!](#)

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Objective

CliniCalm is a B2B SaaS intake quality gate for clinical trial and real-world healthcare data. It validates incoming data automatically at the point of ingestion, flags errors with specific, human-readable reasons, and routes them through a reviewer sign-off workflow with a full audit trail, before anything reaches downstream analytics.

The oncology real-world data space, led by companies like Flatiron Health, has already solved data trust at the dataset level. Flatiron's VALID Framework benchmarks AI-curated variables against historical human-abstractor labels, checks internal consistency, and replicates results against known clinical outcomes, after a dataset is assembled. What's still missing across the industry is a live, per-record decision: for one specific patient's extracted value, right now, is it safe to use without a human checking it. Flatiron's own published prostate cancer research shows two teams defining "progression" two different ways, a live example of exactly the ambiguity that dataset-level validation catches only after the fact.

The Problem

Flatiron's core goals are speed and cost: faster study onboarding, faster evidence generation, more data processed without proportionally more review burden. Two obstacles emerge here:

a. Too much time is spent in manual processing. Every extracted value that isn't obviously trustworthy still needs a human to look at it, and "there's no way to tell which values it actually applies to, versus which are safe to wave through."

b. Errors are uncovered downstream, so even more time is spent on bad data. By the time a mismatched definition or a low-confidence extraction surfaces, in a cohort analysis, in a submission package, it has already been treated as trustworthy for however long it sat unnoticed.

Problem	Reach	Impact	Confidence	Effort	RICE
P1: Manual review and errors burn time and cost	5	5	4	3	33
P2: No live, per-record trust decision	5	5	4	4	25
P3: Lineage stops at the source document	4	4	4	3	21
P4: Definitions drift silently across teams	4	4	3	3	16

P1 is the most visible and costliest problem, but it isn't clear yet what actually causes it. Digging into Flatiron's own published research is where the answer starts to show up: the VALID Framework already proves a whole dataset trustworthy after the fact, and Telescope already traces a value back to its source document. Neither of those operates in real time, on one record, before a human decides whether to trust it. That's P2, and it's the mechanism that P1 is missing. Without it, manual review stays manual because there's no principled way to reduce it, and errors keep surfacing downstream

because nothing catches them earlier. P3 and P4 sharpen the picture further, but P2 is the piece that has to exist first for either of them to matter. Full evidence and reasoning in Appendix A.

Who This Is For

Primary: The Curation Reviewer

- A data curation or RWE team member who currently either trusts an aggregate quality score blindly, or manually reviews far more than necessary.
- Wants to know which values actually need a second look, rather than from scratch.

Secondary: The Data Science / Methodology Owner

- Likely owns or maintains a validation methodology like VALID day to day.
- Needs every accepted value to be defensible under audit

The Solution: CliniCalm

Flatiron already has the beginning of this system, it just stops at the dataset level. VALID validates a whole dataset after curation. Telescope offers source-document traceability for querying. Neither answers the live, per-value question, nor traces a value through the step where trust is most likely to break: the AI extraction itself.

CliniCalm's core is a 9-step intake pipeline — ingest, parse, normalize, validate, flag, review, resolve, revalidate, pass — modeled on a bronze-to-silver data architecture. Raw data lands untouched and immutable in bronze, exactly as the source system sent it, misnamed columns and all; nothing is silently corrected or dropped. The validation gate reads that bronze record, checks it against explicit rules, and only rows that pass — either automatically or after a named human signs off — get promoted to canonical, analysis-ready silver. A flagged row never disappears and never gets guessed at; it sits in quarantine with the rule it violated attached, waiting for human review.

On top of that foundation, this phase adds three things a dataset-level checkpoint like VALID structurally cannot do, because none of them are decidable until you're looking at one specific value, in context, before it's absorbed into an aggregate:

1. **Per-value confidence triage**, benchmarked against historical gold-standard labels, applied live and per-record — not batched or retrospective. A value with a high-confidence match (>95%) against the historical label auto-accepts. A value below that threshold is flagged and always routes to a human; there's no confidence level at which the system decides on its own that a low-confidence value is fine. The point is to stop spending human attention on the values that don't need it, so the review that does happen is concentrated where it matters.
2. **Multi-hop lineage**, tracing a value through every transformation step it passed through, not just back to one source document. A reviewer looking at a flagged value can open the source clinical note, the specific AI extraction model and version that pulled the value out of it, and

the normalization step that mapped it into the canonical schema, and each hop is individually inspectable. This depth is what every adjacent competitor (John Snow Labs' Patient Journey Intelligence, Carta Healthcare's Lighthouse, Komodo's Marmot) stops short of: all of them trace back to a source document, none of them treat the AI extraction step itself as an inspectable hop with its own model, version, and confidence.

3. **Semantic consistency checks**, holding a value back from warehouse load if it conflicts with an existing canonical definition of the same variable — instead of letting two teams' definitions quietly coexist until they surface as a discrepancy months later, the way Flatiron's own published prostate cancer research shows two teams defining "progression" two different ways. The check runs at ingestion, when a conflict is still a five-minute conversation, not an after-the-fact reconciliation.

Guardrails already expected at this scale (volume anomaly detection, schema drift detection, completeness checks against known prevalence) support the core loop but are not the differentiator; they're table stakes every serious intake system needs, and are what CliniCalm includes them for. The live decision layer, the lineage depth, and the semantic check are what a curation reviewer and a methodology owner (see Who This Is For) can't get anywhere else today. Full solution rationale and rejected alternatives in Appendix B. Demonstrable prototype ([try it live](#)) is described in Appendix C.

How We'll Know It's Working

Key Metric: Live Auto-Accept Accuracy Rate

% of values auto-accepted without human review that would be judged correct on the next audit.

At 30 days, watch **Flag Resolution Time**, whether flagged values reach a reviewer and get resolved quickly, or pile up unaddressed. At 60 days, **Live Auto-Accept Accuracy Rate** should rise without **Flag Rate** collapsing to near-zero, meaning the threshold is calibrated. At 90 days, **Semantic Conflict Catch Rate** is the number for stakeholders: if conflicting definitions are being caught before warehouse load instead of surfacing downstream in analysis, the hypothesis holds. Full metrics table in Appendix A.

Timeline

Task	Date
Stakeholder conversation / validation	Day 0
Retrospective test on one already-shipped dataset	+14 days
Prototype refinement based on findings	+21 days
Scoped pilot proposal	+30 days

Top Risks

- **This mechanism is not novel.** John Snow Labs, Carta Healthcare, and Komodo's Marmot all ship close analogs. Mitigated by leading with that fact rather than hiding it, and by naming the two places CliniCalm goes further: lineage through a non-deterministic AI step, and semantic consistency enforced at ingestion.
- **The confidence threshold and the cost of a false-negative flag are unknown from outside any specific customer.** Mitigated by treating calibration as the first real engineering conversation with each target account's data team.
- **It's unclear whether a company like Flatiron has already solved this internally without publishing it.** Mitigated by asking directly rather than assuming the gap is real.

Full risk table in Appendix E.

What's Next

The immediate next step is a scoped pilot: one dataset, rebuilt retrospectively through the decision layer and compared side by side against what shipped originally, before any broader build commitment is asked for.

Beyond that, the same intake-quality-gate pattern applies well past oncology RWE. Optum, Roche, and Exelixis all run large clinical or real-world data operations facing the same underlying tension: reducing study and data onboarding time while keeping data trustworthy enough to defend later. A validated pilot with one customer becomes the reference case for expanding into adjacent health tech and pharma data platforms with the same onboarding-speed-versus-trust problem. Full detail in Appendix F.

Appendix

A. Evidence & Full Metrics

P1 (manual review and cost burden) is evidenced below as the downstream effect of P2–P4, not as an independent claim — see the reasoning in the main Problem section for why P2 is the piece that has to exist first.

P2: No live, per-record trust decision

Evidence: Flatiron's leadership has described the aspiration directly: comparing AI-curated data against historical gold-standard labels while still avoiding a "black box." VALID answers this at the dataset level, retrospectively. Nothing published describes a live, per-record version of the same

decision.

Why it matters: This is the foundational gap.

P3: Lineage stops at the source document

Evidence: Telescope's public description claims source-document traceability. John Snow Labs' Patient Journey Intelligence, Carta Healthcare's Lighthouse, and Komodo's Marmot all provide comparable single-hop provenance. None trace the AI extraction step itself as an inspectable hop with its own confidence and model version attached.

Why it matters: As ambient AI (via partnerships like DeepScribe) moves upstream of extraction, the chain gets a second non-deterministic AI step. Single-hop provenance becomes less sufficient exactly as this becomes more urgent.

P4: Definitions drift silently across teams

Evidence: Flatiron's own published prostate cancer research documents two definitions of "progression" in use — clinician-documented worsening versus a PSA-based threshold — reconciled after the fact rather than prevented at ingestion.

Why it matters: This is a real, named, evidenced failure mode, not a hypothetical one.

Measuring Success

Metric Type	Metric Name	Definition	Why it Matters
Key Metric	Live Auto-Accept Accuracy Rate	% of auto-accepted values still judged correct on later audit	Measures whether the triage threshold is trustworthy, not just fast
Secondary	Flag Resolution Time	Time from flag raised to reviewer sign-off	Shows whether the review workflow is usable under real load
Secondary	Semantic Conflict Catch Rate	% of true definitional conflicts caught before warehouse load vs. discovered downstream	Directly tests the value of the semantic consistency check
Secondary	Lineage Completeness	% of flagged values with a fully inspectable multi-hop chain	Tests whether the lineage differentiator holds up outside a demo dataset

Guardrail Metrics

Metric Type	Metric Name	Definition	Why it Matters
Guardrail	Flag Rate	% of values flagged for human review	A rate near zero signals the threshold is too lenient, not that quality improved
Guardrail	Reviewer Override Rate	% of auto-accepted values a reviewer later disputes	Rising rate signals the threshold needs recalibration

30/60/90 Day Expectations

At 30 days, watch Flag Resolution Time. If flagged values aren't reaching a reviewer promptly, the routing mechanism needs fixing before the triage logic does. At 60 days, Live Auto-Accept Accuracy Rate should be rising without Flag Rate collapsing toward zero — the combination that separates real calibration from a threshold that's just been loosened — and Lineage Completeness should hold near 100%, confirming the lineage differentiator survives outside a demo dataset. By 90 days, Semantic Conflict Catch Rate is the number for stakeholders. If conflicts still aren't being caught pre-ingestion, the next question is whether the canonical-definition registry is too narrow, not whether the concept itself is wrong.

B. Solution Prioritization: RICE

What already exists, and why it isn't yet enough

VALID and Telescope are real, published advances, and this case study doesn't second-guess them. What they don't do is operate at the individual record level, in real time, with a named human accountable for each trust decision. That's the specific space CliniCalm occupies.

Solutions Prioritization: RICE

Solution	Reach	Impact	Confidence	Effort	RICE
Per-value confidence triage	5	5	4	3	33
Governed audit trail / sign-off	4	4	4	2	32
Semantic consistency check at ingestion	4	4	3	3	16
Multi-hop lineage (through the AI step)	4	5	3	4	15
Delegated review (assign flags to other teams)	3	3	3	3	9
Real-time representativeness/bias flagging	3	3	2	4	4.5

The prioritized approach

The triage mechanism and the audit trail score highest on confidence and lowest on effort, they're the load-bearing structure everything else sits on. Lineage and the semantic check score lower on confidence, harder to validate from outside any one company, but are what make CliniCalm distinct from a generic quality dashboard, worth the added effort specifically because no adjacent competitor shows either publicly.

Out of scope for this phase

Real-time representativeness/bias flagging is a natural extension of VALID's existing aggregate bias checks, but is speculative. No direct evidence confirms this is currently unaddressed anywhere, so it's a future direction, not a scoped requirement.

Considered and rejected

A generic third-party data-quality tool instead of a purpose-built gate. Rejected as the sole solution. These enforce schema, type, and range rules well, but don't natively provide the reviewer-assignment, resolution-note, and audit-trail workflow this product's value depends on.

Blocking the entire batch until every flag clears, all-or-nothing. Rejected as too rigid. Some flag types, especially compliance-related ones, should hold the batch, and others shouldn't. That distinction is reflected in Appendix D's blocking-category logic.

Consensus-based sign-off, requiring multiple reviewers to agree, instead of one named signature. Diluting accountability across multiple reviewers undermines the exact audit-trail value proposition the product is built around. The point is one name, decision, and reason.

Fully automated resolution of flagged values without human review. Rejected, since it directly contradicts the trust and transparency premise the whole product is built around.

C. Product Flow (Prototype Walkthrough)

This section documents the interactive prototype built to make the *foundational* mechanism demonstrable in a short conversation. It deliberately scopes to the intake-gate loop — validation, flagging, human sign-off, and audit trail — not the full confidence-triage / lineage / semantic-conflict system described in the Solution above; those are the next build phase, specified as requirements in Appendix D, not yet implemented here.

Stage 1: Run validation. A sample intake batch (18 mock patient-visit records: lab test, value, unit, site) runs through deterministic validation on demand. Every row is checked against four rule types: required-field presence, data type, reference-range, and schema/column-name conformance. Rows that fail any check are flagged with the specific reason, not a generic red flag; e.g. "labValue 'high' is not numeric" or "Column 'site_ID' should be 'siteId'".

Stage 2: Review the findings. Each flagged row opens into a detail view: the raw ingest payload exactly as received (so a misnamed column is visible, not silently normalized away), a reviewer-assignment field, and a required resolution note — sign-off is disabled until both are present. An operational audit trail logs every action against the finding as it happens: the rule that failed and on which field, who it was assigned to, and who signed off with what justification.

Stage 3: Sign-off and rollup. Once signed off, the row is marked passed and the batch-level dashboard (total rows, flagged, resolved, average time to resolution) updates live. A companion "How

it works" view walks a non-technical audience through the underlying 9-step pipeline and bronze-to-silver framing conceptually.

Deliberately out of this build: per-value confidence scoring and auto-accept, multi-hop lineage across the AI-extraction step, semantic-conflict detection against a canonical definition registry, delegation to a named team, and a batch-level authenticated sign-off with a blocking compliance-hold category. These remain scoped as requirements in Appendix D and are the subject of the Phase 2 retrospective validation described in Appendix F.

Wireframes

#	Screen	What it shows	Maps to
1	Home page, pre-validation	Initial load, nothing run yet	Stage 0
2	Flagged batch view	Live-demo table right after "Run validation" – 18 rows, mixed Passed/Flagged badges, each flagged row's specific reason inline	Stage 1
3	Raw payload & review panel	Finding modal open, "raw ingest payload" expanded (shows a misnamed column exactly as received), reviewer dropdown + required note field, both empty	Stage 2 (traceability + gate)
4	Signed off, audit trail	Same finding after sign-off – compact confirmation banner + the 3-event audit trail (created → assigned → signed off) with real timestamps	Stage 2 (governed sign-off – #2 highest-RICE solution)
5	Live rollup	Summary dashboard tiles after a couple of findings are resolved – Total/Flagged/Resolved/Avg time all updated	Stage 3
6	How it works (companion)	Pipeline diagram + 9-step workflow grid	Non-technical explainer, not a feature demo

Screen 1

Screen 2

CC

CliniCalm

Automated intake validation for clinical trial data

Reset demoRun validation

Live demoHow it works

TOTAL ROWS
18
in this intake batch

FLAGGED
run validation to check

RESOLVED
closed by reviewers

AVG. TIME TO RESOLUTION
simulated, for demo pacing

Intake batch — clinical_intake_2026-01.csv

AllFlaggedResolvedPassed

PATIENT ID	VISIT DATE	LAB TEST	VALUE	UNIT	SITE	STATUS	ISSUE	REVIEWER	ACTION
P001	2026-01-05	Hemoglobin	14.2	g/dL	S01	Not run	—	—	—
P002	2026-01-06	Glucose	95	mg/dL	S01	Not run	—	—	—
P003	2026-01-06	Potassium	4.1	mmol/L	S02	Not run	—	—	—
P004	2026-01-07	Hemoglobin	13.8	g/dL	S02	Not run	—	—	—
P005	2026-01-07	Glucose	high	mg/dL	S01	Not run	—	—	—
P006	2026-01-08	Potassium	4.4	mmol/L	S03	Not run	—	—	—
P007	2026-01-08	Hemoglobin	—	g/dL	S01	Not run	—	—	—
P008	2026-01-09	Glucose	118	mg/dL	S02	Not run	—	—	—
P009	2026-01-09	Potassium	9.8	mmol/L	S03	Not run	—	—	—
P010	2026-01-10	Hemoglobin	15.1	g/dL	S03	Not run	—	—	—

Screen 3

CC

CliniCalm

Automated intake validation for clinical trial data

Reset demoRe-run validation

Live demoHow it works

TOTAL ROWS
18
in this intake batch

FLAGGED
6
awaiting review

RESOLVED
0
of 6 flagged

AVG. TIME TO RESOLUTION
simulated, for demo pacing

Intake batch — clinical_intake_2026-01.csv

AllFlaggedResolvedPassed

PATIENT ID	VISIT DATE	LAB TEST	VALUE	UNIT	SITE	STATUS	ISSUE	REVIEWER	ACTION
P001	2026-01-05	Hemoglobin	14.2	g/dL	S01	Passed	No issues found	—	No action needed
P002	2026-01-06	Glucose	95	mg/dL	S01	Passed	No issues found	—	No action needed
P003	2026-01-06	Potassium	4.1	mmol/L	S02	Passed	No issues found	—	No action needed
P004	2026-01-07	Hemoglobin	13.8	g/dL	S02	Passed	No issues found	—	No action needed
P005	2026-01-07	Glucose	high	mg/dL	S01	Flagged	labValue "high" is not numeric Invalid data type	Unassigned	In queue 15m Review
P006	2026-01-08	Potassium	4.4	mmol/L	S03	Passed	No issues found	—	No action needed
P007	2026-01-08	Hemoglobin	—	g/dL	S01	Flagged	"labValue" is required but missing Missing field	Unassigned	In queue 2h 35m Review
P008	2026-01-09	Glucose	118	mg/dL	S02	Passed	No issues found	—	No action needed
P009	2026-01-09	Potassium	9.8	mmol/L	S03	Flagged	Potassium 9.8 mmol/L is outside normal range (3.5-5.1 mmol/L) Out-of-range value	Unassigned	In queue 57m Review
P010	2026-01-10	Hemoglobin	15.1	g/dL	S03	Passed	No issues found	—	No action needed
P011	2026-01-11	Glucose	88	mg/dL	S01	Passed	No issues found	—	No action needed

Screen 4

CC

CliniCalm

Automated intake validation for clinical trial data

Reset demoRe-run validation

Live demoHow it works

TOTAL ROWS
18
in this intake batch

FLAGGED
6
awaiting review

RESOLVED
0
of 6 flagged

AVG. TIME TO RESOLUTION
simulated, for demo pacing

Intake batch — clinical_intake_2026-01.csv

AllFlaggedResolvedPassed

PATIENT ID	VISIT DATE	LAB TEST	VALUE	UNIT	SITE	STATUS	ISSUE	REVIEWER	ACTION
P001	2026-01-05	Hemoglobin	14.2	g/dL	S01	Passed	No issues found	—	No action needed
P002	2026-01-06	Glucose	95	mg/dL	S01	Passed	No issues found	—	No action needed
P003	2026-01-06	Potassium	4.1	mmol/L	S02	Passed	No issues found	—	No action needed
P004	2026-01-07	Hemoglobin	13.8	g/dL	S02	Passed	No issues found	—	No action needed
P005	2026-01-07	Glucose	high	mg/dL	S01	Flagged	labValue "high" is not numeric Invalid data type	Unassigned	In queue 15m Review
P006	2026-01-08	Potassium	4.4	mmol/L	S03	Passed	No issues found	—	No action needed
P007	2026-01-08	Hemoglobin	—	g/dL	S01	Flagged	"labValue" is required but missing Missing field	Unassigned	In queue 2h 35m Review
P008	2026-01-09	Glucose	118	mg/dL	S02	Passed	No issues found	—	No action needed
P009	2026-01-09	Potassium	9.8	mmol/L	S03	Flagged	Potassium 9.8 mmol/L is outside normal range (3.5-5.1 mmol/L) Out-of-range value	Unassigned	In queue 57m Review
P010	2026-01-10	Hemoglobin	15.1	g/dL	S03	Passed	No issues found	—	No action needed
P011	2026-01-11	Glucose	88	mg/dL	S01	Passed	No issues found	—	No action needed

P013 — Schema mismatch

Validation details & raw ingest payload

```
{  "patientID": "P013",  "visitDate": "2026-01-11",  "labTest": "Hemoglobin",  "labValue": 12.9,  "unit": "g/dL",  "site_ID": "S02"}
```

REVIEW & SIGN-OFF

Assigned reviewer

Select a reviewer...

Resolution note *REQUIRED

What was wrong, and how was it fixed?

Resolved & passed

STATUS: Flagged

OPERATIONAL AUDIT TRAIL

- Finding created
 - Action: ClinCalm Intake Gate — Deterministic rule FIELD_NAME_MATCH failed on field "site_ID"

Finding ID: F2ND-CN813

Close

Screen 5

CC

CliniCalm

Automated intake validation for clinical trial data

Reset demoRe-run validation

Live demoHow it works

TOTAL ROWS
18
in this intake batch

FLAGGED
6
awaiting review

RESOLVED
0
of 6 flagged

AVG. TIME TO RESOLUTION
simulated, for demo pacing

Intake batch — clinical_intake_2026-01.csv

AllFlaggedResolvedPassed

PATIENT ID	VISIT DATE	LAB TEST	VALUE	UNIT	SITE	STATUS	ISSUE	REVIEWER	ACTION
P001	2026-01-05	Hemoglobin	14.2	g/dL	S01	Passed	No issues found	—	No action needed
P002	2026-01-06	Glucose	95	mg/dL	S01	Passed	No issues found	—	No action needed
P003	2026-01-06	Potassium	4.1	mmol/L	S02	Passed	No issues found	—	No action needed
P004	2026-01-07	Hemoglobin	13.8	g/dL	S02	Passed	No issues found	—	No action needed
P005	2026-01-07	Glucose	high	mg/dL	S01	Flagged	labValue "high" is not numeric Invalid data type	Unassigned	In queue 15m Review
P006	2026-01-08	Potassium	4.4	mmol/L	S03	Passed	No issues found	—	No action needed
P007	2026-01-08	Hemoglobin	—	g/dL	S01	Flagged	"labValue" is required but missing Missing field	Unassigned	In queue 2h 35m Review
P008	2026-01-09	Glucose	118	mg/dL	S02	Passed	No issues found	—	No action needed
P009	2026-01-09	Potassium	9.8	mmol/L	S03	Flagged	Potassium 9.8 mmol/L is outside normal range (3.5-5.1 mmol/L) Out-of-range value	Unassigned	In queue 57m Review
P010	2026-01-10	Hemoglobin	15.1	g/dL	S03	Passed	No issues found	—	No action needed
P011	2026-01-11	Glucose	88	mg/dL	S01	Passed	No issues found	—	No action needed

P013 — Schema mismatch

Validation details & raw ingest payload

REVIEW & SIGN-OFF

Assigned reviewer

Priya Nair (Clinical Data Coordinator)

Resolution note *REQUIRED

Confirmed with site: column arrived as "site_ID" instead of "siteID", value is S01.

Resolved & passed

Resolved at: 05/24/2026 19:20

STATUS: Resolved

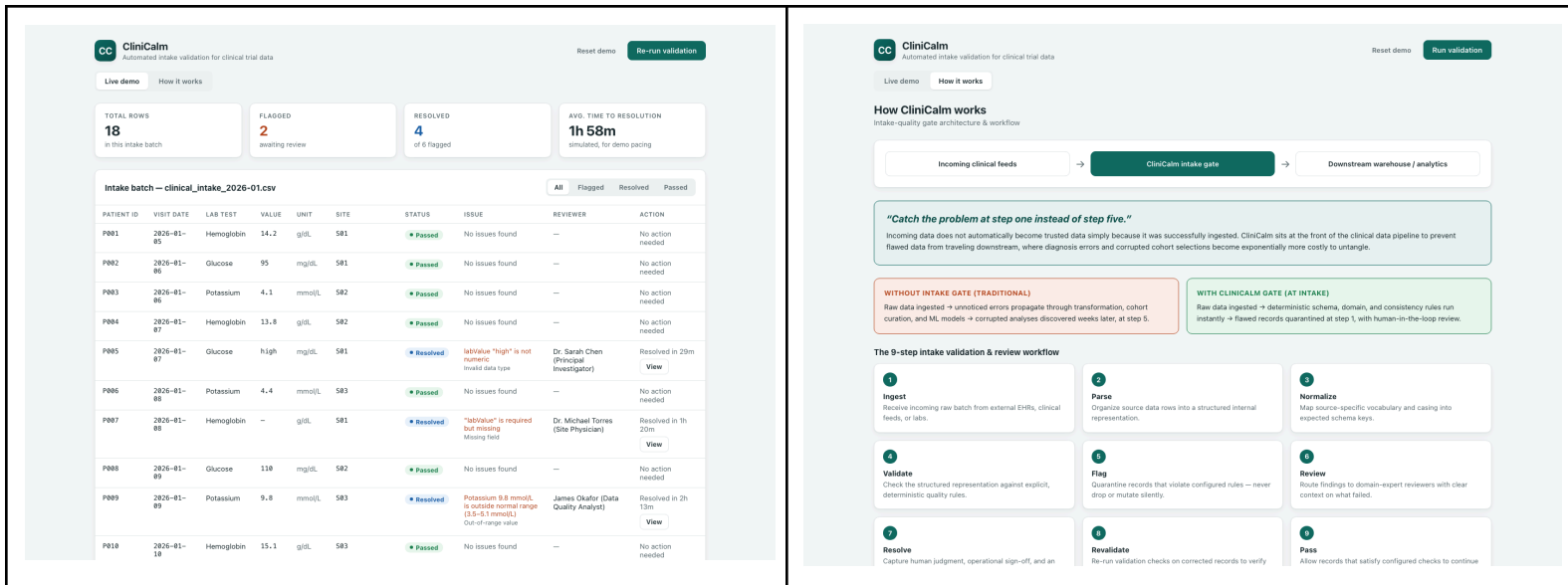
OPERATIONAL AUDIT TRAIL

- Finding created
 - Action: ClinCalm Intake Gate — Deterministic rule FIELD_NAME_MATCH failed on field "site_ID"
- Assigned to Priya Nair (Clinical Data Coordinator)
 - Action: Clinical Data Operations Lead — Quarantined record routed to Priya Nair (Clinical Data Coordinator)
- Resolution signed off — Passed
 - Action: Priya Nair (Clinical Data Coordinator) — Signed off with note: "Confirmed with site: column arrived as "site_ID" instead of "siteID", value is S01."

Finding ID: F2ND-CN813

Close

Screen 6



D. User Stories, Acceptance Criteria, Edge Cases

D1. Prototype: Acceptance Criteria (what Appendix C actually demonstrates)

- Every row is checked against four rule types (required field, data type, reference range, schema / column-name match); a failure produces a specific, human-readable reason, not a bare pass/fail.
- A flagged row's raw ingest payload is viewable exactly as received, misnamed columns included.
- Sign-off on a finding requires both an assigned reviewer and a written resolution note; sign-off action stays disabled until both exist.
- Every action on a finding (created, assigned, signed off) writes to that finding's audit trail with an actor and a timestamp.
- The dashboard tiles and status filter (All / Flagged / Resolved / Passed) update live as validation runs and findings are resolved.

D2. Target System — User Stories & Requirements (the proposed CliniCalm from the Solution section, not yet built)

User Stories

- As a curation reviewer, I want to see a confidence score per extracted value, so I know which ones I actually need to look at.
- As a curation reviewer, I want to trace a low-confidence value through every step it passed through, including the AI extraction itself, so I'm not trusting a black box.
- As a data science or methodology owner, I want conflicting definitions of the same variable caught before they enter the warehouse, so I'm not reconciling a discrepancy months later.
- As a reviewer facing a flag outside my expertise, I want to delegate it to the right team and pick up where I left off later.

- As someone defending this data under audit, I want a permanent record of who reviewed or resolved each flagged value, and why.

Requirements

- Every extracted value receives a confidence score; values above a defined threshold auto-accept, values below are flagged, with no exceptions.
- A flagged value's full lineage is inspectable, not summarized to a single citation.
- A semantic conflict between two definitions of the same variable is held back from warehouse load until reconciled.
- A flag can be assigned to a specific team, status is visibly distinct, persists across sessions.
- Sign-off is unavailable until every flag is reviewed or resolved via delegation, except a flag assigned to a blocking category eg. Compliance; must clear regardless of other flags' status.

Edge Cases (target system)

- A flag is delegated but never resolved: state persists indefinitely rather than silently expiring.
- Reviewer unchecks previous flag: sign-off, if already unlocked, re-locks immediately.
- Two flags are both assigned to the same team: each tracked independently.
- A value has no confident extraction at all, not just low confidence: shown as a real, visible flagged value rather than a blank placeholder.

E. Key Event Tracking Sheet & Assumptions/Risks

Key Event Tracking Sheet

Event Name	Description	Trigger	Supports
Batch Scored	All values in a batch receive confidence scores	System run	Flag Rate (denominator)
Flag Raised	A value or pipeline check falls below threshold	System response	Live Auto-Accept Accuracy Rate, Flag Rate
Flag Delegated	A flag is assigned to a specific team	Button click	Flag Resolution Time (alternate path)
Flag Resolved	A flag is confirmed reviewed or resolved by delegation	Checkbox / confirmation	Flag Resolution Time, Semantic Conflict Catch Rate
Batch Signed Off	Named reviewer signs the batch	Button click	Live Auto-Accept Accuracy Rate (retrospective input)

Assumptions & Risks

Assumption	Risk	Mitigation	Threshold to Act	Detection Metric
A confidence threshold can be calibrated per variable	A single global threshold hides differences between variable types	Treat per-variable calibration as the first real engineering conversation with each customer's data team	Explicitly deferred to a real pilot	Reviewer Override Rate
A target account hasn't already solved this internally	The gap is only real if it's actually unaddressed	Ask directly rather than assume	If a customer confirms an internal equivalent exists, the pitch reframes around what's still missing from it	Direct stakeholder conversation
Delegation to another team meaningfully speeds resolution vs. one reviewer	Delegation could become a way to indefinitely defer a flag	Distinguish blocking categories like Compliance that cannot be soft-bypassed regardless of delegation status	If delegated flags show materially longer resolution times, revisit the delegation model	Flag Resolution Time, segmented by resolution path

F. Dependencies & Future Steps

Open Questions

- Who inside a target account like Flatiron actually owns this pain point, a data curation team, a research sciences team, or a compliance function?
- Has a company like Flatiron already solved this internally without publishing it?
- What's the actual cost of a false-negative flag versus a false-positive one? Unknown without real internal data from a design partner.

Infrastructure Requirements

- Access to a customer's own historical gold-standard labels, to benchmark live per-record confidence scoring against
- A canonical definition registry for variables prone to drift, to power semantic consistency
- Real database versioning capabilities to underpin the audit trail

Internal Dependencies

- A design partner willing to validate or refute the core gap directly
- Data science input on realistic confidence-threshold calibration per variable type

- One already-shipped dataset from a design partner, to run the retrospective validation test proposed above

Future Steps

Phase 2: Retrospective validation with a design partner Rebuild one already-shipped dataset through the decision layer and compare it side by side against what shipped originally. The low-risk test proposed as the very next step, before any broader build commitment.

Phase 3: Scoped pilot Depends on Phase 2 showing a meaningful gap between what shipped and what the decision layer would have caught.

Phase 4: Expansion beyond oncology RWE Optum, Roche, and Exelixis all run large clinical or real-world data operations wrestling with the same tension: reducing onboarding and review time while keeping data defensible. Once validated with one design partner, the same intake-quality-gate pattern becomes the pitch for these adjacent accounts, each facing a version of the same trade-off between speed and trust.

Phase 5: Self-service canonical-definition registry The semantic consistency check currently implies engineering hardcodes each variable's definition. A real product needs methodology owners to define, own, and version these themselves, without a build ticket every time a new variable or edge case comes up.

Phase 6: Integration with existing lineage and catalog tooling Rather than building lineage storage from scratch, explore integrating with tools a design partner may already run, such as Unity Catalog or OpenLineage, once their actual stack is known. Could meaningfully cut build effort versus a fully custom lineage layer.

Phase 7: API and webhook layer So a customer's own orchestration (Airflow, dbt, or similar) can trigger and consume CliniCalm's decisions programmatically, not only through a manual UI. Necessary for real production adoption beyond a single reviewer working in a browser tab.

Not phased, lowest priority overall Flagging possible bias in real time, as data comes in. A natural next step from the aggregate bias checks that already exist, but there's no evidence yet that this is actually missing anywhere.